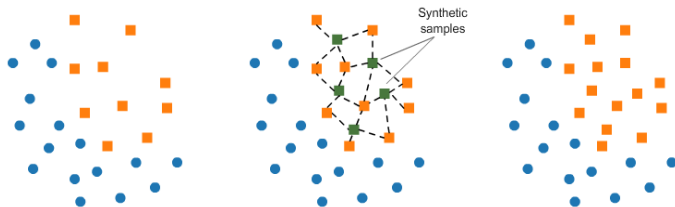
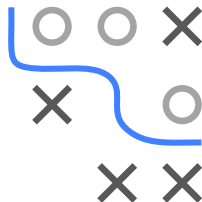


OVERSAMPLING: SMOTE

- SMOTE creates **synthetic instances** of minority class.
- Interpolate between neighboring minority instances.
- Instances are created in $\mathcal{X}$ rather than in $\mathcal{X} \times \mathcal{Y}$.
- Algorithm: For each minority class instance:
 - Find its k nearest minority neighbors.
 - Randomly select one of these neighbors.
 - Randomly generate new instances along the lines connecting the minority example and its selected neighbor.



SMOTE: GENERATING NEW EXAMPLES

- Let $\mathbf{x}^{(i)}$ be the feature of the minority instance and let $\mathbf{x}^{(j)}$ be its nearest neighbor. The line connecting the two instances is

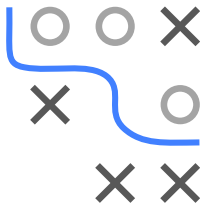
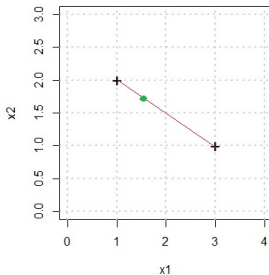
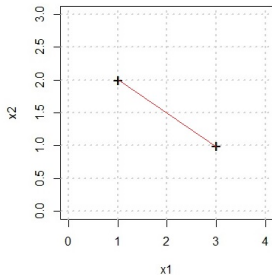
$$(1 - \lambda)\mathbf{x}^{(i)} + \lambda\mathbf{x}^{(j)} = \mathbf{x}^{(i)} + \lambda(\mathbf{x}^{(j)} - \mathbf{x}^{(i)})$$

where $\lambda \in [0, 1]$.

- By sampling a $\lambda \in [0, 1]$, say $\tilde{\lambda}$, we create a new instance

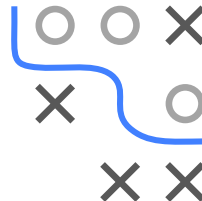
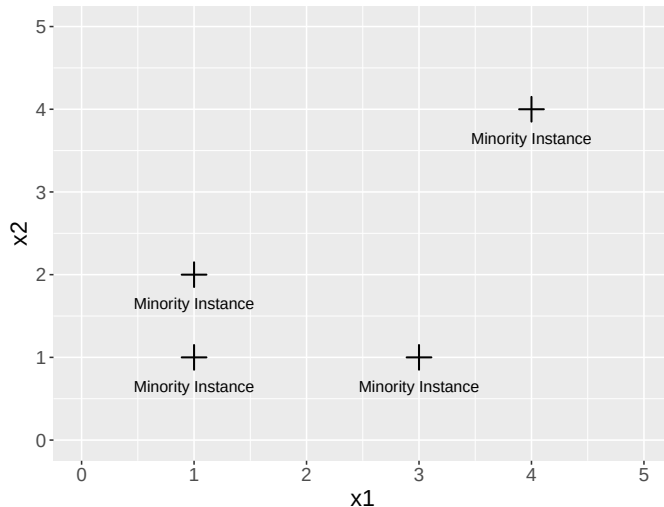
$$\tilde{\mathbf{x}}^{(i)} = \mathbf{x}^{(i)} + \tilde{\lambda}(\mathbf{x}^{(j)} - \mathbf{x}^{(i)})$$

Example: Let $\mathbf{x}^{(i)} = (1, 2)^\top$ and $\mathbf{x}^{(j)} = (3, 1)^\top$. Assume $\tilde{\lambda} \approx 0.25$.



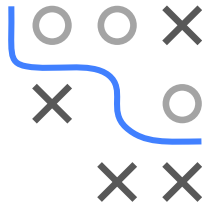
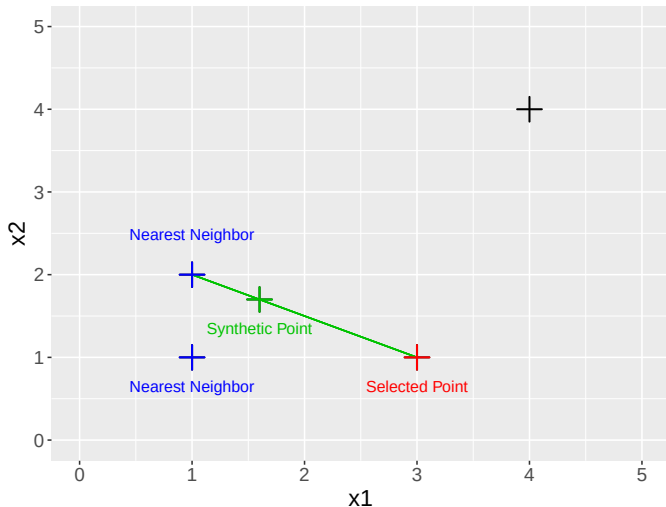
SMOTE: VISUALIZATION

For an imbalanced data situation, take four instances of the minority class. Let $K = 2$ be the number of nearest neighbors.



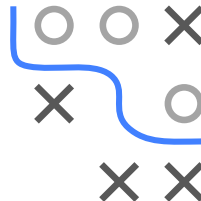
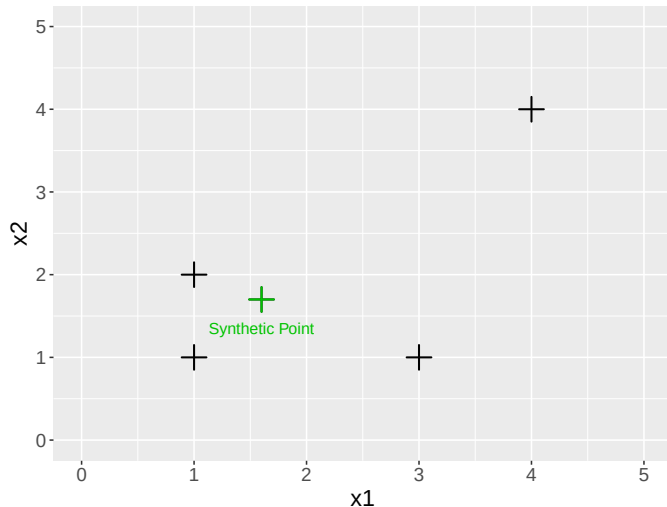
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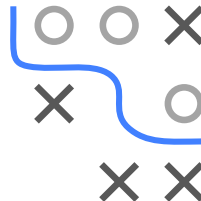
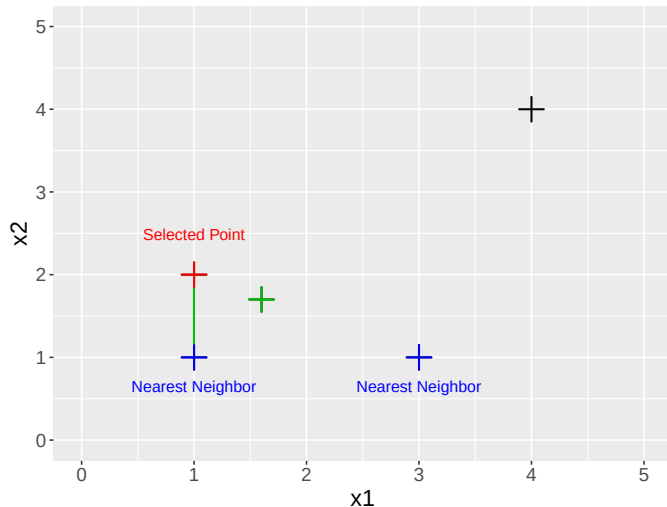
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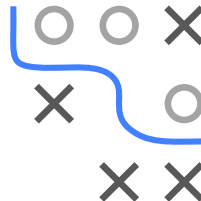
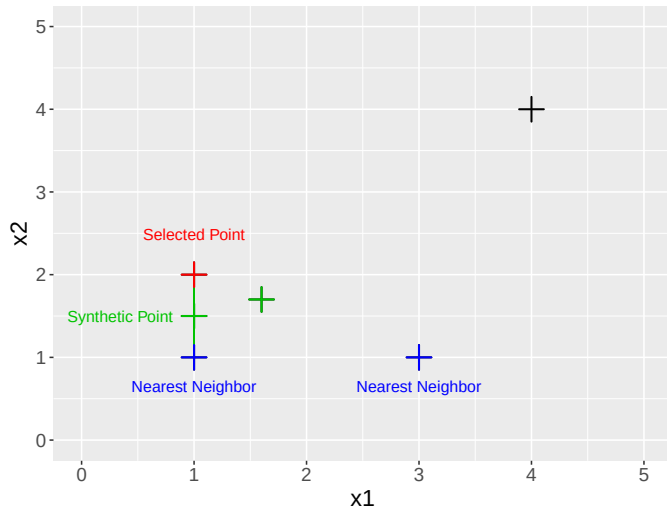
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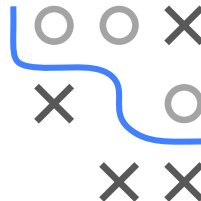
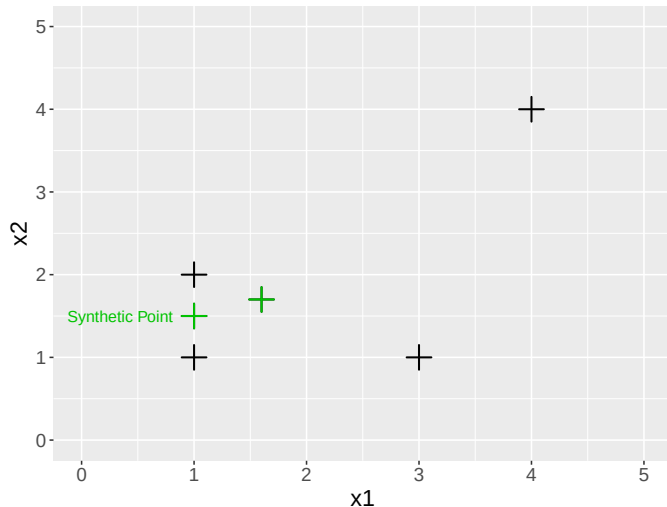
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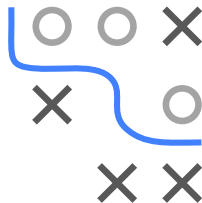
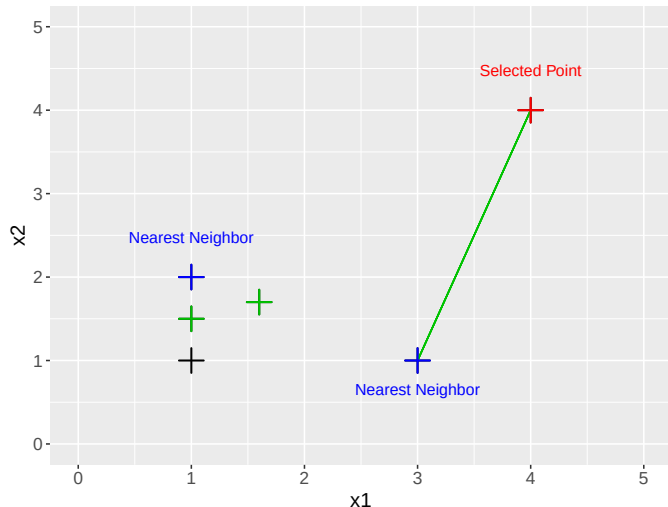
SMOTE: VISUALIZATION

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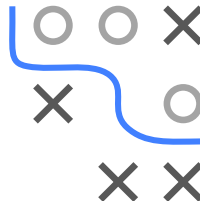
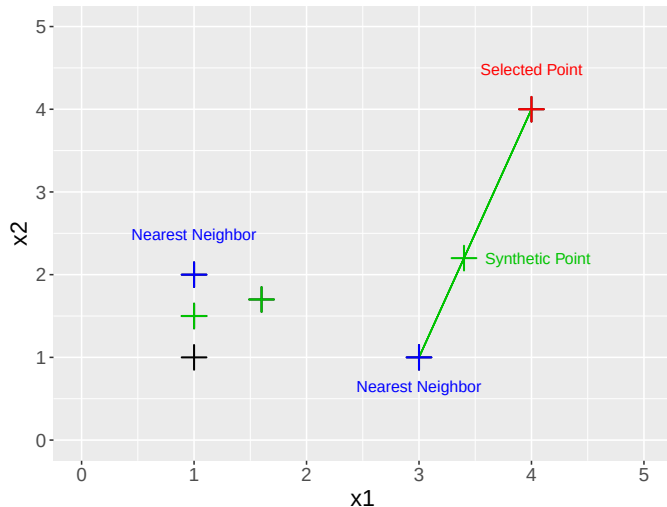
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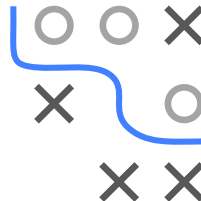
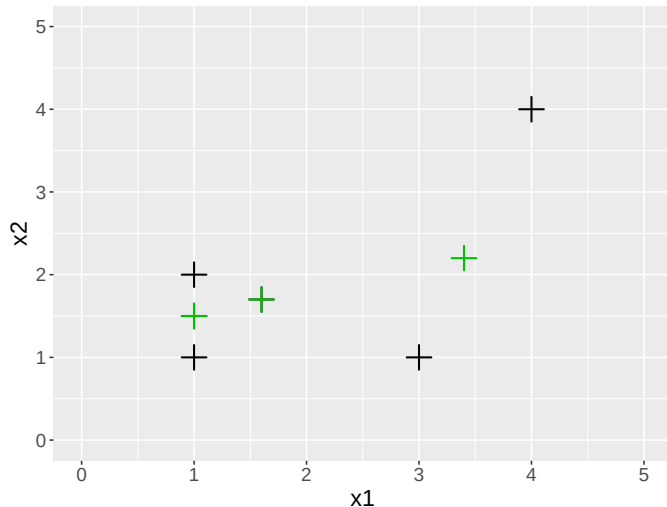
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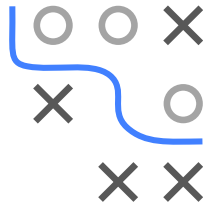
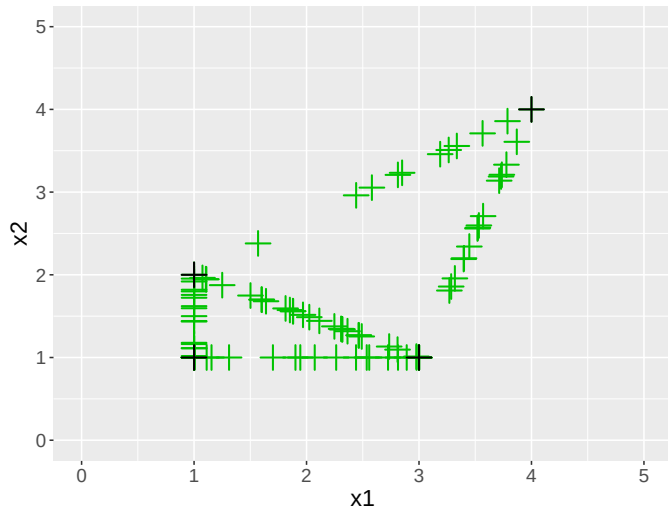
SMOTE: VISUALIZATION

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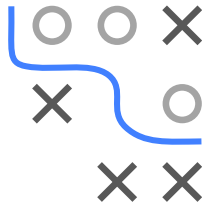
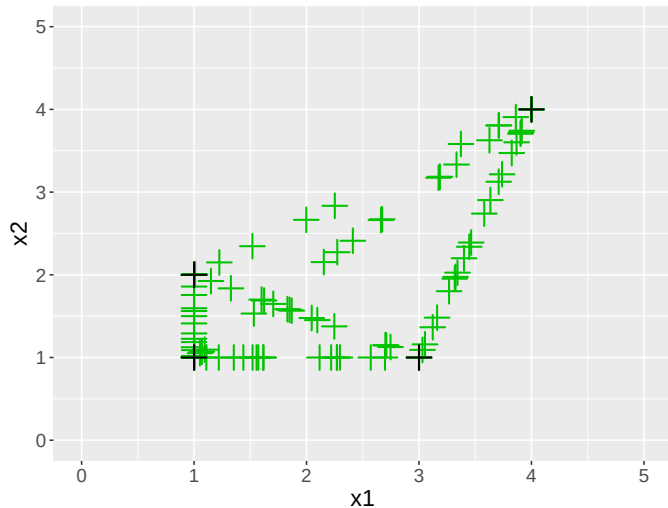
SMOTE: VISUALIZATION CONTINUED

After 100 iterations of SMOTE for $K = 2$ we get:



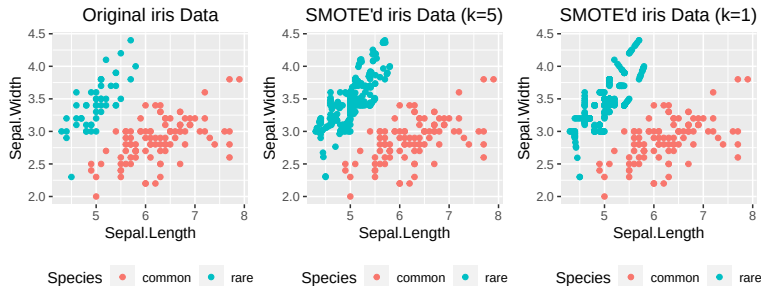
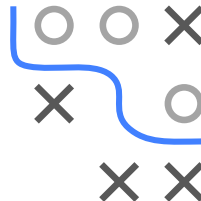
SMOTE: VISUALIZATION CONTINUED

After 100 iterations of SMOTE for $K = 3$ we get:



SMOTE: EXAMPLE

- Iris data set with 3 classes and 50 instances per class.
- Make the data set “imbalanced”:
 - relabel one class as positive
 - relabel two other classes as negative



SMOTE enriches minority class feature space.

SMOTE: DIS-/ADVANTAGES

- Generalize decision region for minority class instead of making it quite specific, such as by random oversampling.
- Well-performed among the oversampling techniques and is the basis for many oversampling methods: Borderline-SMOTE, LN-SMOTE, . . . (over 90 extensions!)
- Prone to overgeneralizing as it pays no attention to majority class.

